

AI 모델 성능 개선 결과

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Parallel LSTM 성능 개선 결과

학습데이터

- 관심 수위 : 궁내교 (2) 대곡교 (3.8)

- smoothing : 궁내교 (1) 대곡교 (2)

- learning_rate = 0.0001 / 0.0003 / 0.0005

- Loss_function = MSE / MAE / MAPE

- Seq_length = 36 (6시간) / 48 / 60 / 72

- Num_of_layer = 6 / 12 (각 LSTM 3개 처리) / 18 / 36

- 궁내교 영향 없는 관측소 및 컬럼 제외 / 전체 강우

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1412개 데이터 파일 학습에 사용 (결측 5개 + 검증 3개)

검증 : 학습 데이터에서 제외한 3개 + 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : ~ 2022 68번 (988개)

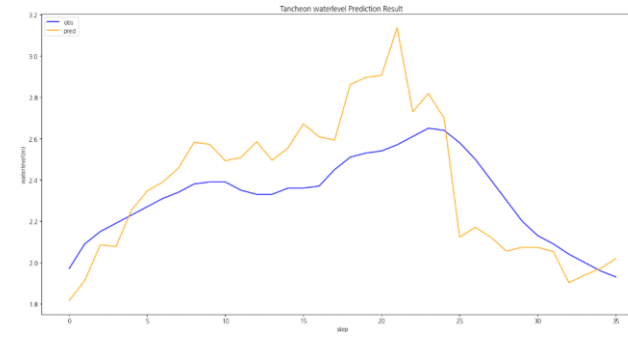
- 검증 : ~ 2024 96번 (282개)

- 테스트 : ~ 2025 109번 (142개)

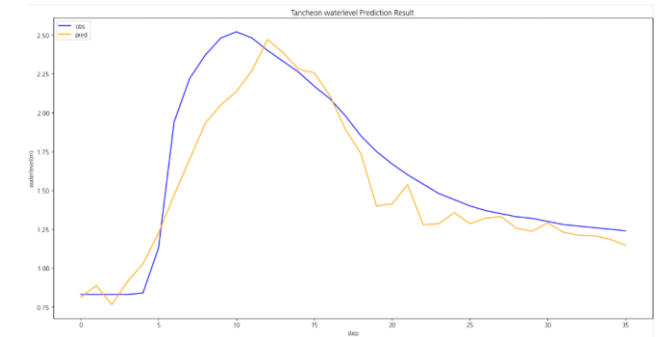
num_of_layer = 18 loss_function = MAE, learning_rate = 0.0005

예측 코드 - **궁내교**

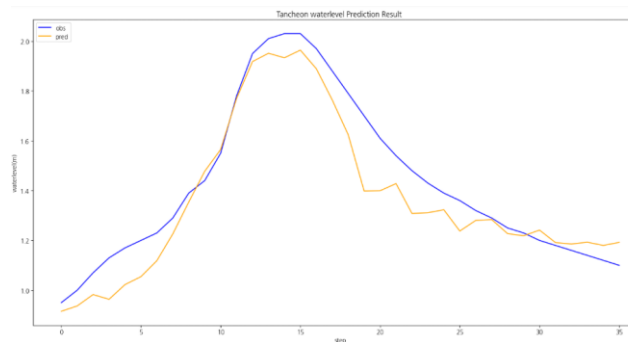
궁내교	
testdata_220712	r2 : -0.2437 mse : 0.0493 mae : 0.1828 rmse : 0.2221
testdata_230711	r2 : 0.8500 mse : 0.0419 mae : 0.1468 rmse : 0.2048
증설 전	
testdata_250716	r2 : 0.8895 mse : 0.0110 mae : 0.0833 rmse : 0.1052
증설 후	
testdata_250814	r2 : 0.7838 mse : 0.0152 mae : 0.0962 rmse : 0.1234



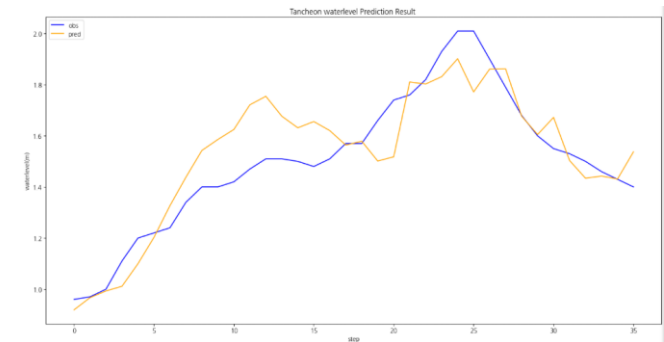
testdata_220712



testdata_230711



testdata_250716

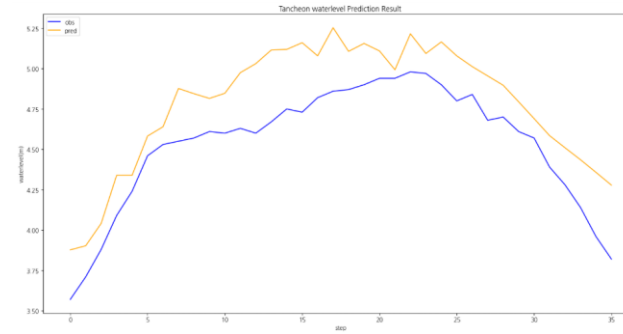


testdata_250814

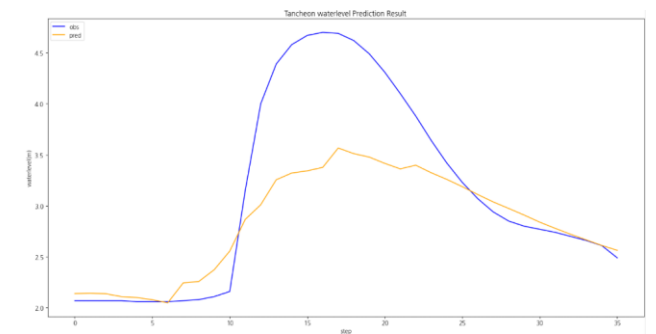
num_of_layer = 18, loss_function = MSE, learning_rate = 0.0001

예측 코드 - 대곡교

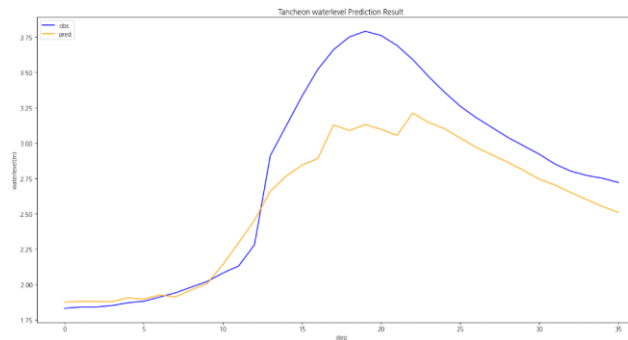
	대곡교
증설 전	testdata_220712 r2 : 0.4649 mse : 0.0749 mae : 0.2530 rmse : 0.2737
	testdata_230711 r2 : 0.6025 mse : 0.3617 mae : 0.3919 rmse : 0.6014
증설 후	testdata_250716 r2 : 0.7837 mse : 0.0998 mae : 0.2374 rmse : 0.3159
	testdata_250814 r2 : 0.6523 mse : 0.1206 mae : 0.2747 rmse : 0.3473



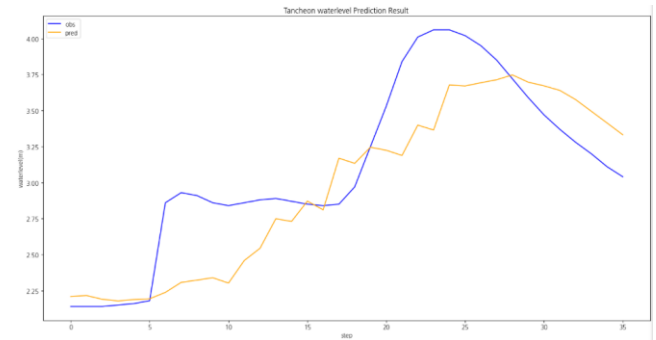
testdata_220712



testdata_230711



testdata_250716

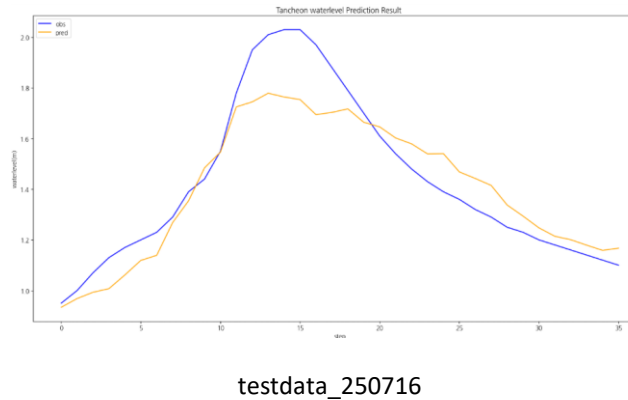
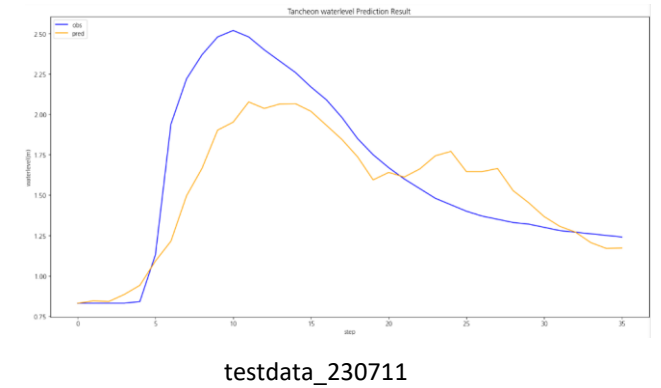
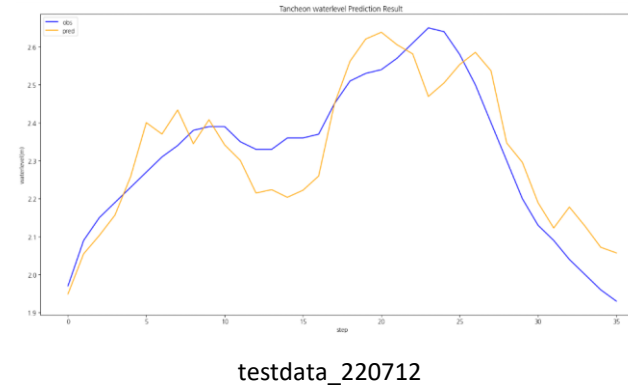


testdata_250814

num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005

예측 코드 - **궁내교**

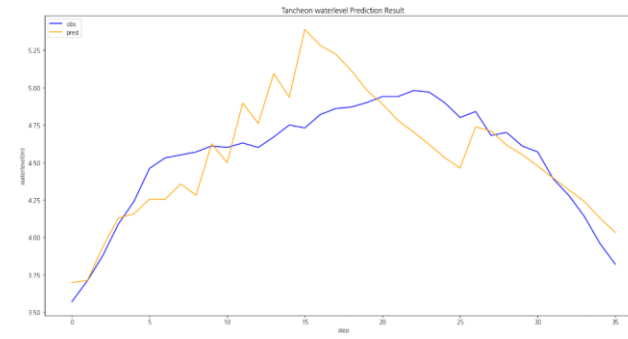
	궁내교
증설 전	testdata_220712 r2 : 0.7886 mse : 0.0083 mae : 0.07876 rmse : 0.0915 testdata_230711 r2 : 0.6794 mse : 0.0896 mae : 0.2132 rmse : 0.2994
증설 후	testdata_250716 r2 : 0.8537 mse : 0.0146 mae : 0.0960 rmse : 0.1211 testdata_250814 r2 : 0.7326 mse : 0.0188 mae : 0.0974 rmse : 0.1372



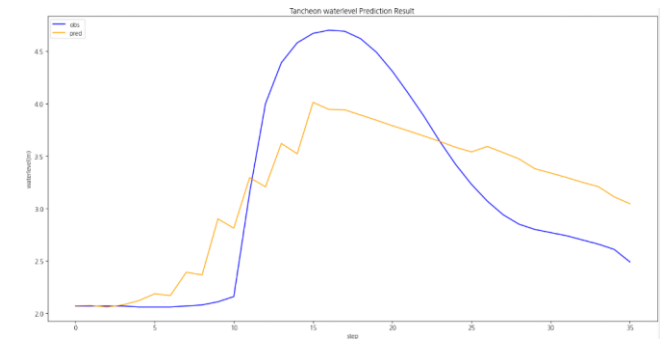
num_of_layer = 18 , loss_function = MAE, learning_rate = 0.0005

예측 코드 - 대곡교

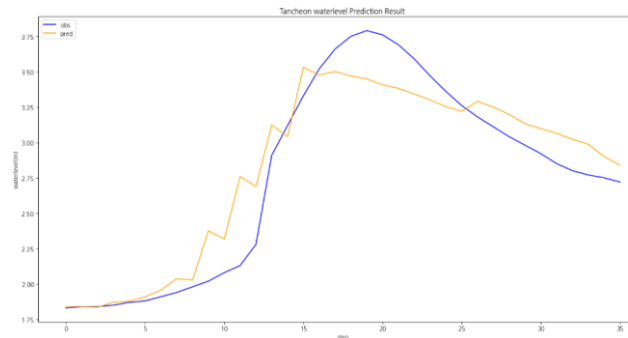
	대곡교
증설 전	testdata_220712 r2 : 0.5988 mse : 0.0561 mae : 0.1851 rmse : 0.2370
	testdata_230711 r2 : 0.6995 mse : 0.2734 mae : 0.4393 rmse : 0.5229
증설 후	testdata_250716 r2 : 0.8985 mse : 0.0468 mae : 0.1692 rmse : 0.2164
	testdata_250814 r2 : 0.6443 mse : 0.1234 mae : 0.2827 rmse : 0.3513



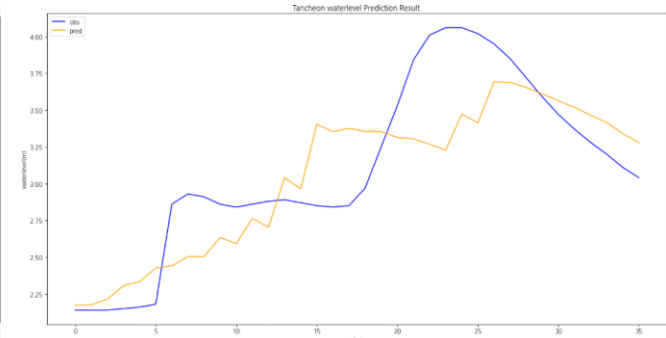
testdata_220712



testdata_230711



testdata_250716

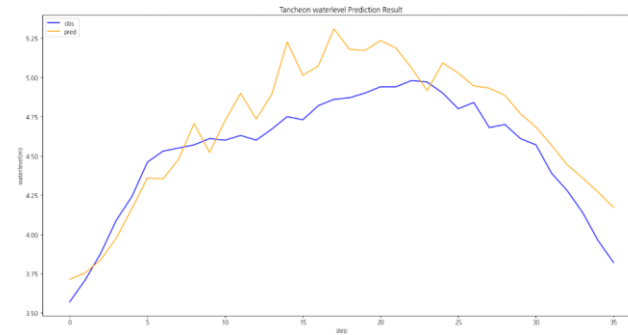


testdata_250814

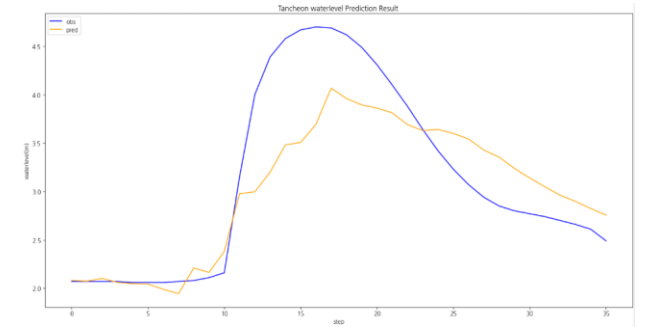
num_of_layer = 12 , loss_function = MSE, learning_rate = 0.0001

예측 코드 - 대곡교

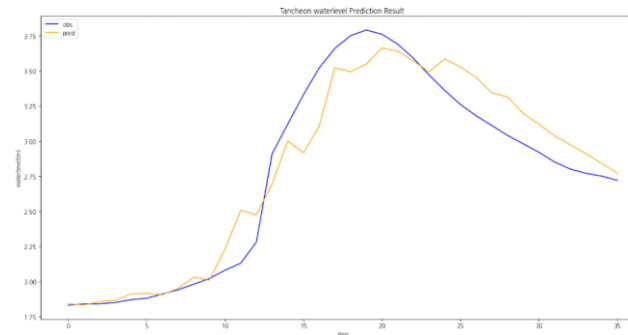
대곡교	
testdata_220712	r2 : 0.6571 mse : 0.0480 mae : 0.1920 rmse : 0.2191
testdata_230711	r2 : 0.7198 mse : 0.2550 mae : 0.3688 rmse : 0.5049
증설 전	
testdata_250716	r2 : 0.9231 mse : 0.0355 mae : 0.1457 rmse : 0.1884
증설 후	
testdata_250814	r2 : 0.8274 mse : 0.0598 mae : 0.2138 rmse : 0.2446



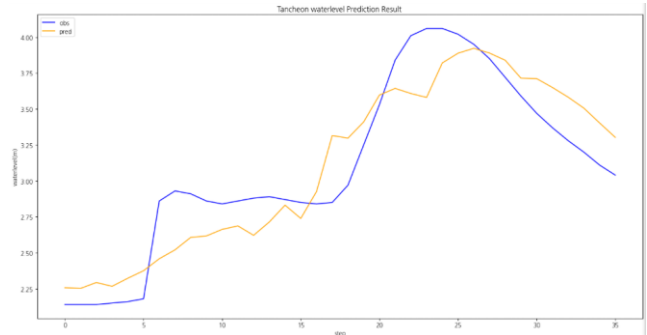
testdata_220712



testdata_230711



testdata_250716



testdata_250814

결론

전체적으로 기존 모델에 비해 전체적으로 낮은 성능,
220712이 명확한 약점

단일 LSTM 개발 결과

학습데이터

- `loss_function` = MSE / MAE / Huber / 피크에 가중치를 두는 MSE
 - * (Huber : 작은 오차는 MSE처럼 , 큰 오차는 MAE처럼)
 - * (MSE , 스케일된 값이 0.7 이상이면 가중치를 3배)
- `learning_rate` = 0.0003 / 0.0001
- `dropout` = 0.0 / 0.1 / 0.01
- `seq_length` = 36
- `Dense` = 1 / 12 / 36 (1번 처리)
- 학습 이벤트 순서 `shuffle`

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

학습 : 총 1420개 中 1412개 데이터 파일 학습에 사용 (결측 5개 + 검증 3개)

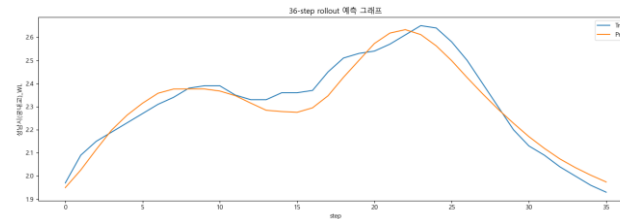
검증 : 학습 데이터에서 제외한 3개 + 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 70% (학습 이벤트 순서 섞어서)
- 검증 : 20%
- 테스트 : 나머지 (10%)

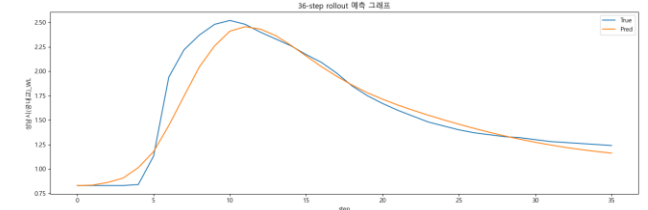
예측 코드 - 궁내교

→ 전체 평균은 좋지만, 250716 이 명확한 약점

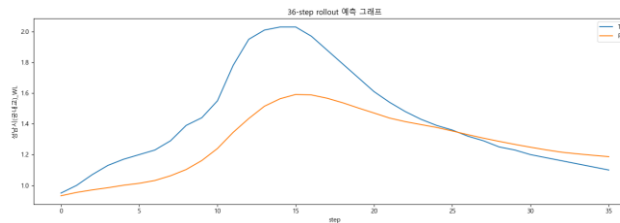
	궁내교
증설 전	testdata_220712 r2 : 0.9379 mse : 0.0025 mae : 0.0427 rmse : 0.0496
	testdata_230711 r2 : 0.9279 mse : 0.0202 mae : 0.0818 rmse : 0.1420
증설 후	testdata_250716 r2 : 0.4562 mse : 0.0546 mae : 0.1751 rmse : 0.2336
	testdata_250814 r2 : 0.8408 mse : 0.0112 mae : 0.0797 rmse : 0.1060



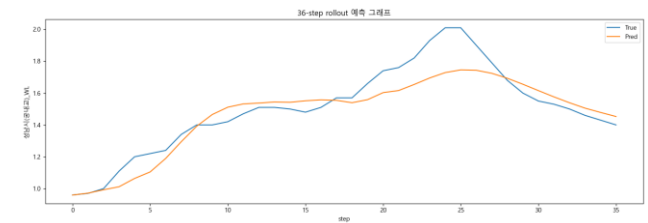
testdata_220712



testdata_230711



testdata_250716

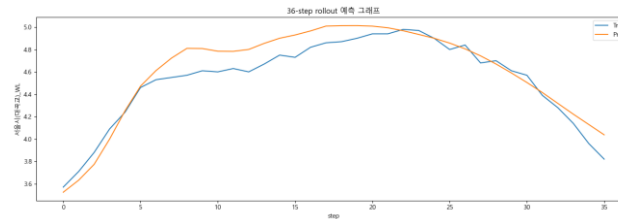


testdata_250814

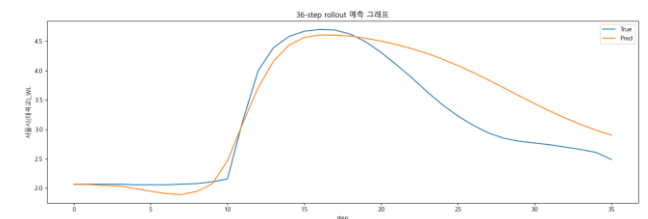
예측 코드 - 대곡교

→ 전체 평균은 좋지만, 250716 이 명확한 약점

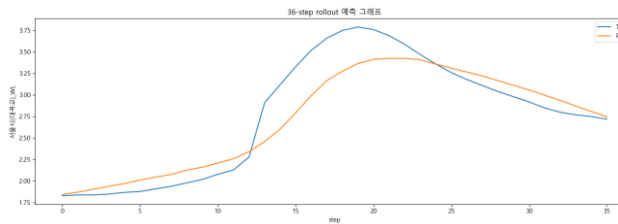
		대곡교
증설 전	testdata_220712	r2 : 0.8927 mse : 0.0150 mae : 0.1010 rmse : 0.1226
	testdata_230711	r2 : 0.7866 mse : 0.1942 mae : 0.3283 rmse : 0.4407
증설 후	testdata_250716	r2 : 0.8683 mse : 0.0608 mae : 0.1862 rmse : 0.2466
	testdata_250814	r2 : 0.8509 mse : 0.0518 mae : 0.1702 rmse : 0.2275



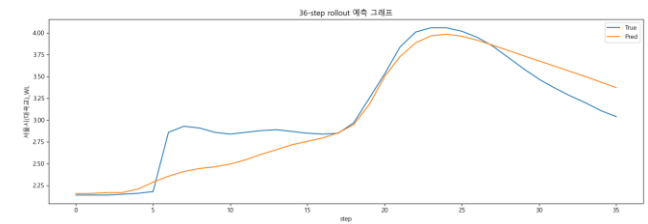
testdata_220712



testdata_230711



testdata_250716

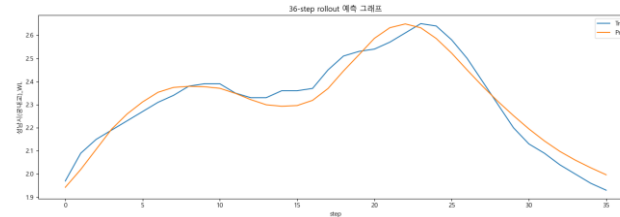


testdata_250814

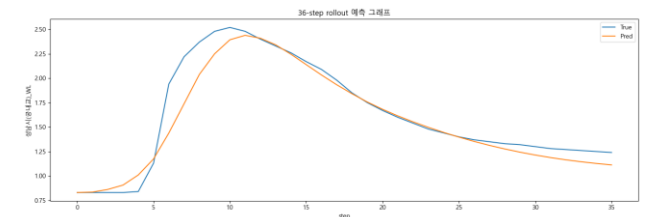
+ 고수위 구간에 가중치를 더 주는 MSE 예측 코드 - **궁내교**

→ 최고 R2 (220712 , 250814)
→ 피크 성능 가장 좋음
→ 250716 번은 아직 약점

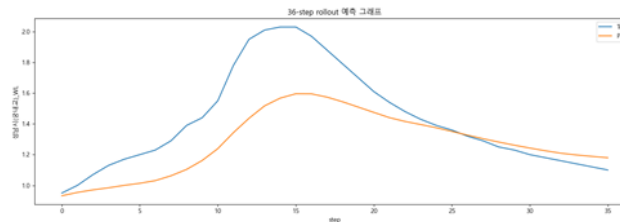
	궁내교
증설 전	
testdata_220712	r2 : 0.9442 mse : 0.0022 mae : 0.0415 rmse : 0.0471
testdata_230711	r2 : 0.9215 mse : 0.0220 mae : 0.0867 rmse : 0.1482
증설 후	
testdata_250716	r2 : 0.4650 mse : 0.0537 mae : 0.1724 rmse : 0.2317
testdata_250814	r2 : 0.8604 mse : 0.0098 mae : 0.0795 rmse : 0.0992



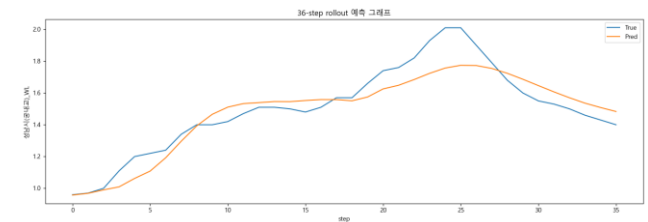
testdata_220712



testdata_230711



testdata_250716



testdata_250814

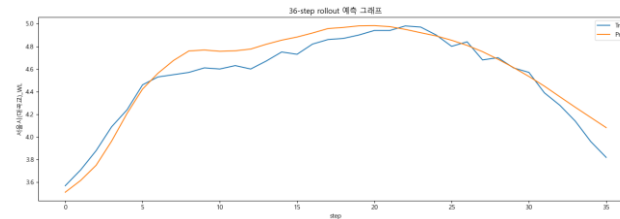
+ 고수위 구간에 가중치를 더 주는 MSE 예측 코드 - 대곡교

→ 최고 R2 (220712 , 250814)
→ 250814 개선 폭이 가장 좋음
→ 250716 은 아직 약점

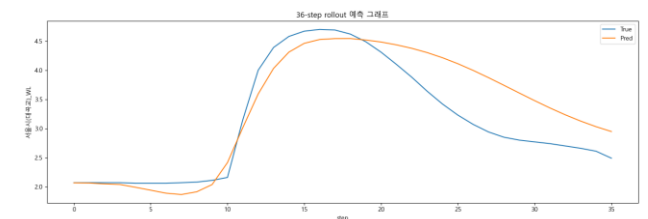
증설 전

증설 후

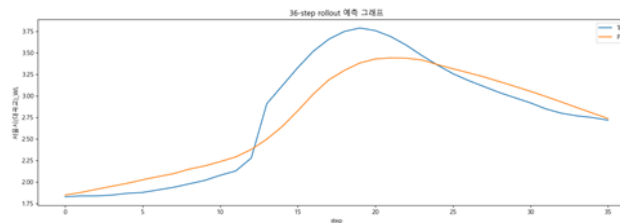
	대곡교
testdata_220712	r2 : 0.9121 mse : 0.0123 mae : 0.0924 rmse : 0.1109
testdata_230711	r2 : 0.7603 mse : 0.2182 mae : 0.3618 rmse : 0.4671
testdata_250716	r2 : 0.8780 mse : 0.0563 mae : 0.1852 rmse : 0.2373
testdata_250814	r2 : 0.8497 mse : 0.0522 mae : 0.1663 rmse : 0.2284



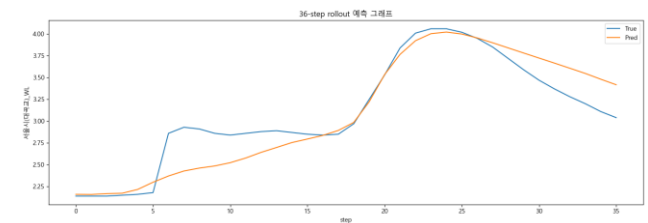
testdata_220712



testdata_230711



testdata_250716



testdata_250814

결론

Parallel LSTM 에 비해 높은 성능,
궁내교는 250716이 명확한 약점,
대곡교는 230711이 명확한 약점